



# 人工智能

让计算机或机器能够执行通常需要人类智能才能完成的任务的整体技术和理论体系。

## 机器学习

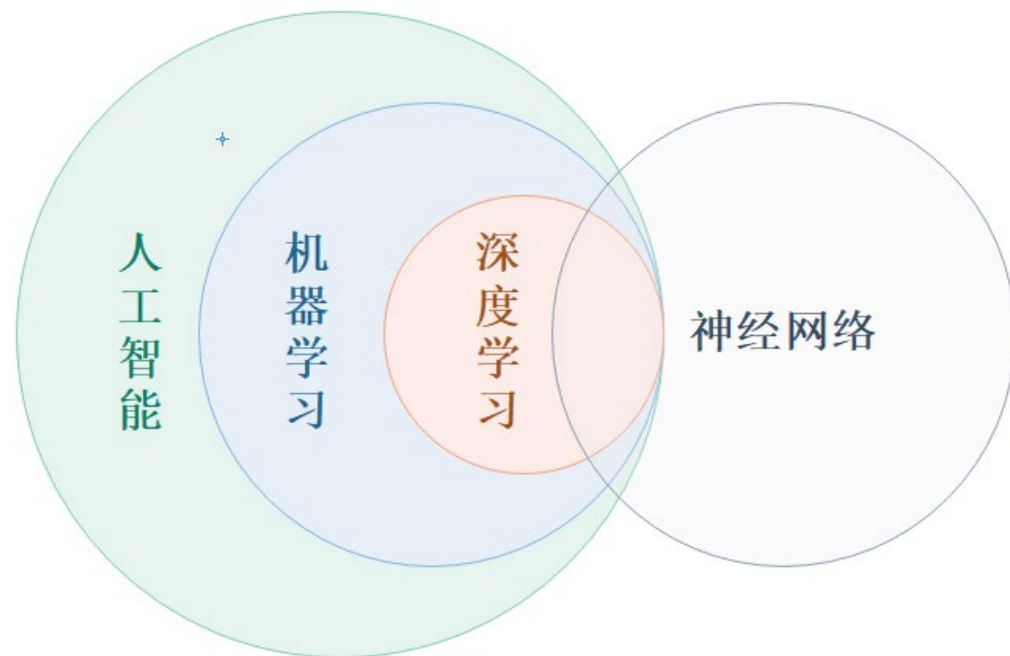
通过从数据中自动学习规律并改进性能，而无需显式编程的算法与方法。

## 深度学习

使用深层架构的机器学习方法，其核心思想是通过多层次的非线性变换，从数据中自动提取更高层次的抽象特征。

## 神经网络

最初是指大脑神经元，人工智能受神经网络的启发，发展出了人工神经网络。



# 人工智能的方法

## Methodologies of Artificial Intelligence

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## Methodologies of Artificial Intelligence

### ◆ Rule-based

- Implemented by direct programming
- Inspired by human heuristics

# 人工智能的方法

## Methodologies of Artificial Intelligence

### ◆ Rule-based

- Implemented by direct programming
- Inspired by human heuristics

### ◆ Data-based

- Expert systems

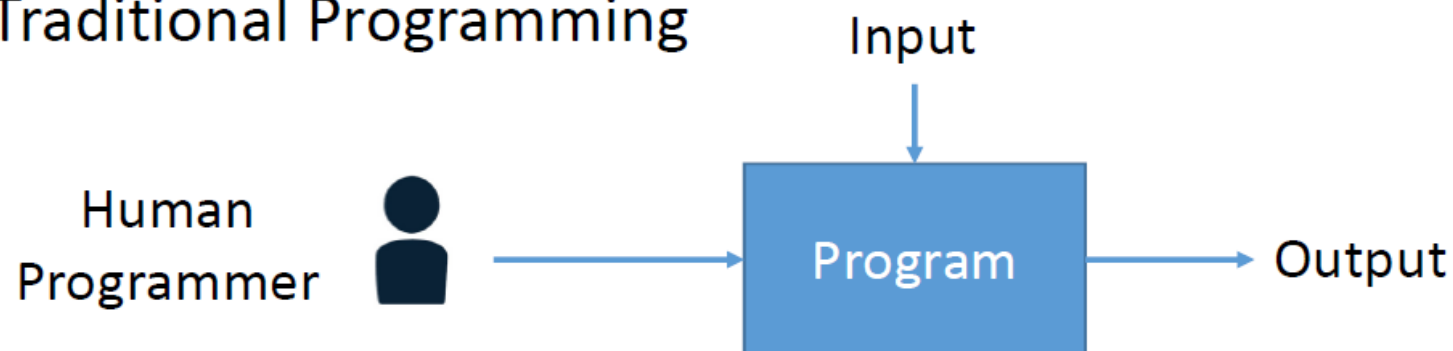
Experts or statisticians create rules of predicting or decision making based on the data

- Machine learning
  - Direct making prediction or decisions based on the data
  - Data Science

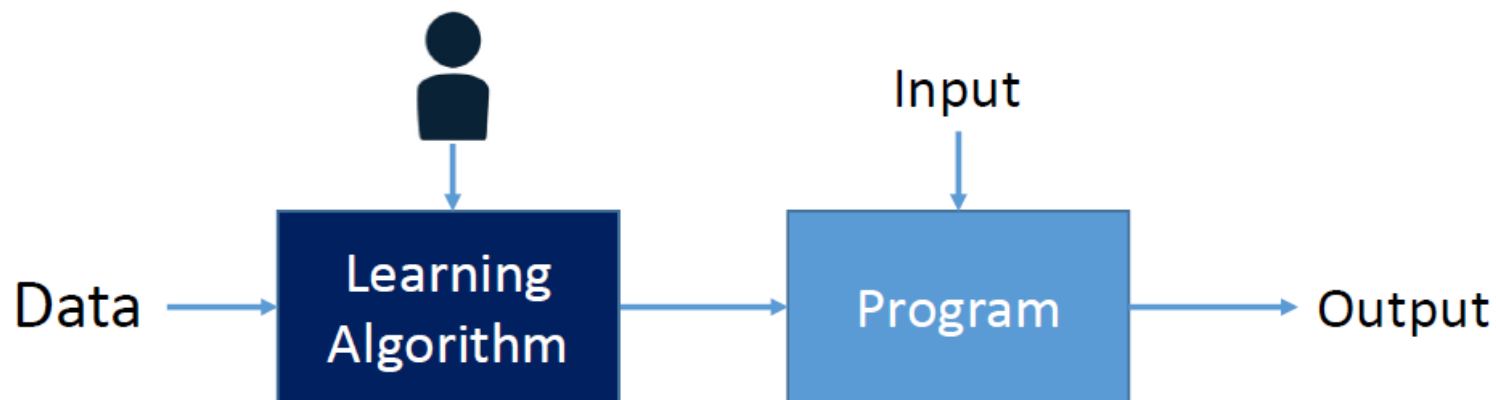
# 机器学习和传统编程的不同

## Traditional Programming vs. Machine Learning

- Traditional Programming



- Machine Learning



# 机器学习适用于... Learning is used when

- Develop systems that are too difficult/expensive to construct manually because they require specific detailed skills or knowledge tuned to a specific task (Speech / face recognition)
- Even if we had a good idea about how to do it, the program might be horrendously complicated. (Robot arm, autonomous helicopter, handwriting recognition, most of natural language processing, game of Go)
- Human expertise does not exist (navigating on Mars)
- Rapid decisions that humans cannot do (high-frequency trading)
- Solution changes in time (routing on a computer network)

# 机器学习适用于... Learning is used when

- Develop systems that can automatically adapt and customize themselves to individual users in a massive scale.
  - Personalized news or mail filter
  - Personalized tutoring
  - Product recommendation
- Discover new knowledge from large databases (data mining).
  - Market basket analysis (e.g. web click data)
  - Medical text mining (e.g. migraines to calcium channel blockers to magnesium)



# 机器学习定义

## What is Machine Learning?

- Learning is any process by which a system improves performance from experience

--- Herbert Simon



Turing Award (1975)

artificial intelligence, the psychology of human cognition

Nobel Prize in Economics (1978)

decision-making process within economic organizations

# 机器学习定义

## What is Machine Learning?

- A more mathematical definition by Tom Mitchell
- Machine learning is the study of algorithms that
  - improvement their performance  $P$
  - at some task  $T$
  - based on experience  $E$
  - with non-explicit programming
- A well-defined learning task is given by  $\langle P, T, E \rangle$

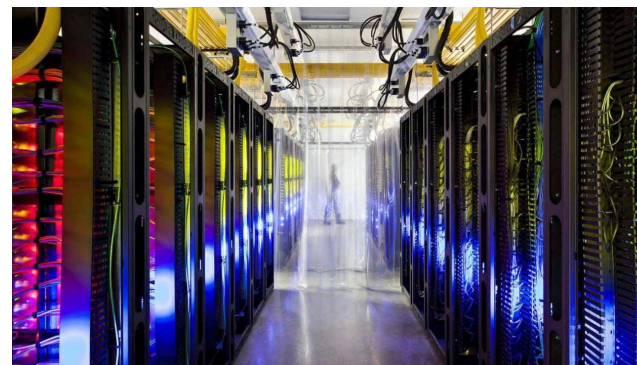
# 机器学习条件

## Why Study Machine Learning? The Time is Ripe

- Many basic effective and efficient algorithms available.
- Large amounts of on-line data available.
- Large amounts of computational resources available.



一万亿参数神经网络



模拟训练一次仅电费大约1000万美元

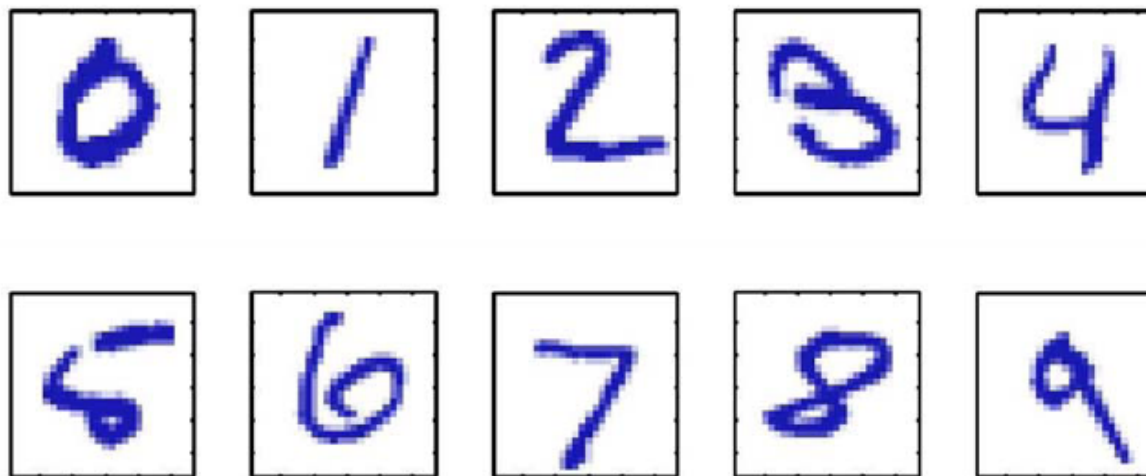
# 学什么 What is Learning?

- Machine learning is the study of algorithms that improvement their performance  $P$  at some task  $T$  based on experience  $E$  with non-explicit programming
- What is the task?
  - Classification
  - Regression
  - Clustering
  - Dimensionality reduction

# 学什么 What is Learning?

## Example 1: hand-written digit recognition

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Images are 28 x 28 pixels

Represent input image as a vector  $\mathbf{x} \in \mathbb{R}^{784}$

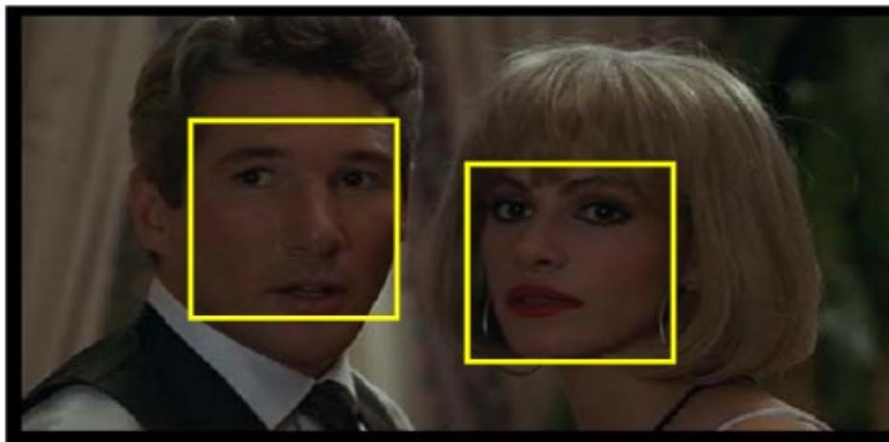
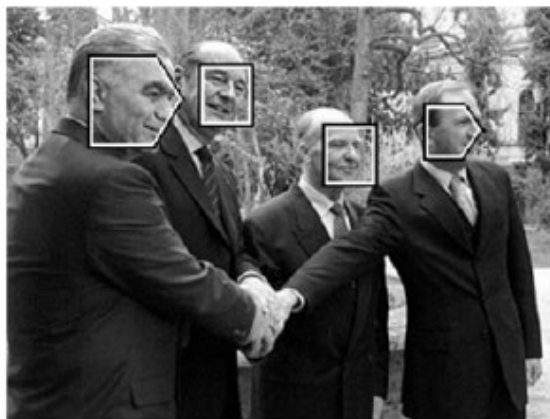
Learn a classifier  $f(\mathbf{x})$  such that,

$$f : \mathbf{x} \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

# 学什么

## What is Learning?

### Example 2: Face detection



- Again, a supervised classification problem
- Need to classify an image window into three classes:
  - non-face
  - frontal-face
  - profile-face

# 学什么 What is Learning?

Classifier is learnt from labelled data

---

## Training data for frontal faces

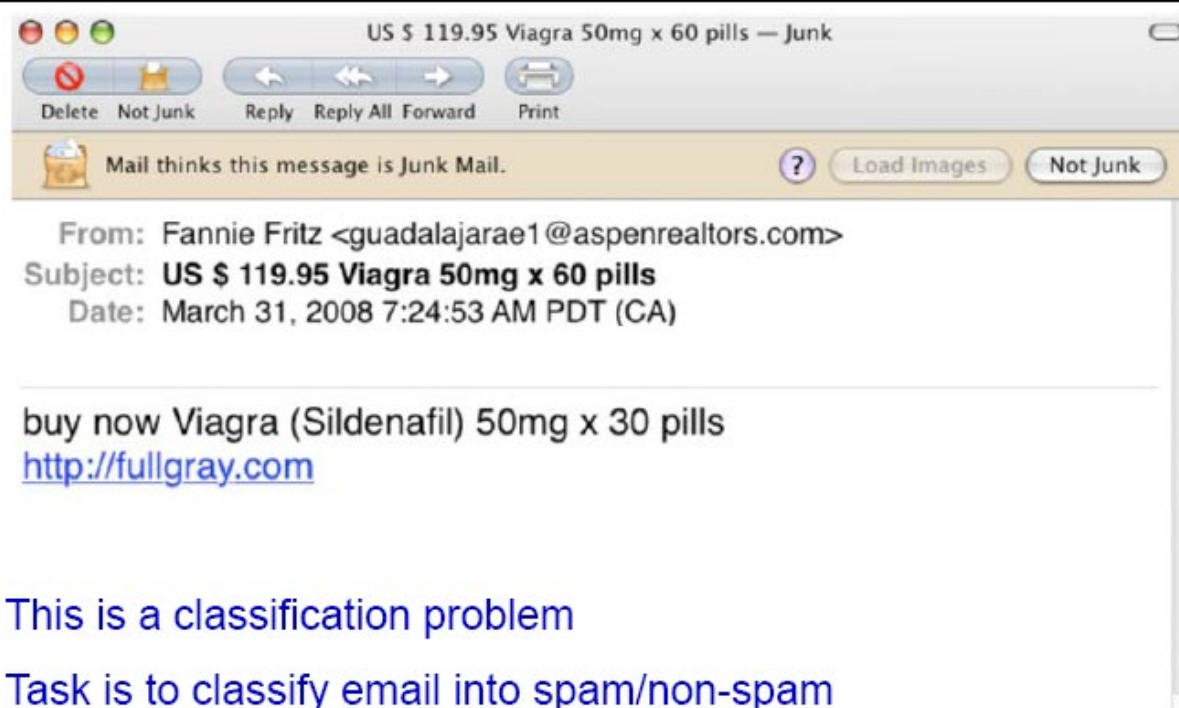
- 5000 faces
  - All near frontal
  - Age, race, gender, lighting
- $10^8$  non faces
- faces are normalized
  - scale, translation





# 学什么 What is Learning?

## Example 3: Spam detection



- This is a classification problem
- Task is to classify email into spam/non-spam
- Data  $x_i$  is word count, e.g. of viagra, outperform, “you may be surprized to be contacted” ...
- Requires a learning system as “enemy” keeps innovating



# 学什么 What is Learning?

## Example 4: Stock price prediction



- Task is to predict stock price at future date
- This is a regression task, as the output is continuous

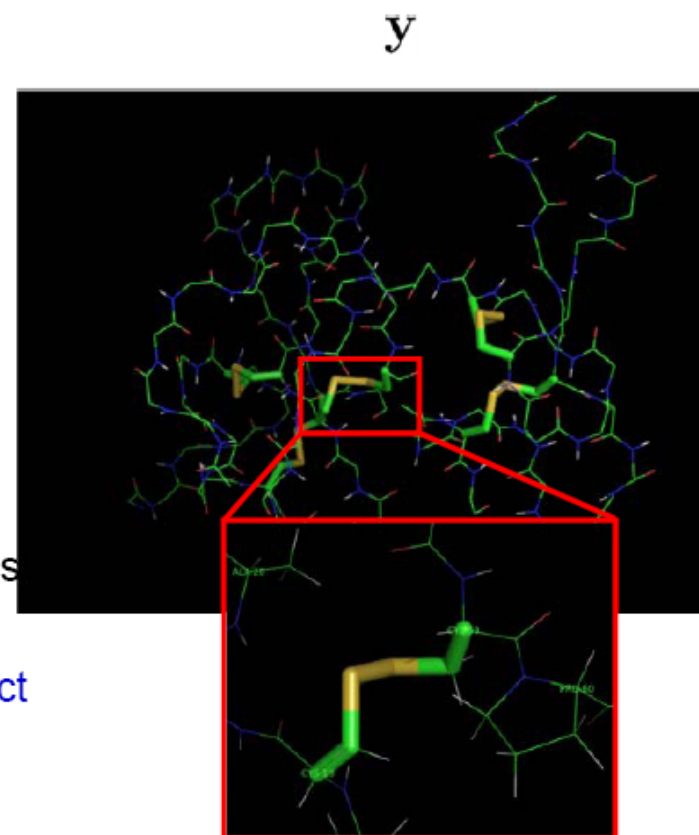
# 学什么 What is Learning?

## Example 5: Computational biology

x

```

AVITGACERDLQCG
KGTCCA VSLWIKSV
RVCTPVGTSGEDCH
PASHKIPFSGQRMH
HTCPCAPNLACVQT
SPKKFKCLSK
    
```



Protein Structure and Disulfide Bridges

Regression task: given sequence predict  
3D structure

Protein: 1IMT

# 学什么 What is Learning?

## Web examples: Recommender systems

### People who bought Hastie ...

#### Frequently Bought Together

Customers buy this book with [Pattern Recognition and Machine Learning \(Information Science and Statistics\) \(Information Science and Statistics\)](#) by Christopher M. Bishop



+



Price For Both: **£104.95**

[Add both to Basket](#)

#### Customers Who Bought This Item Also Bought

Page 1



[Pattern Recognition and Machine Learning \(Information Science and Statistics\)](#) by Christopher M. Bishop  
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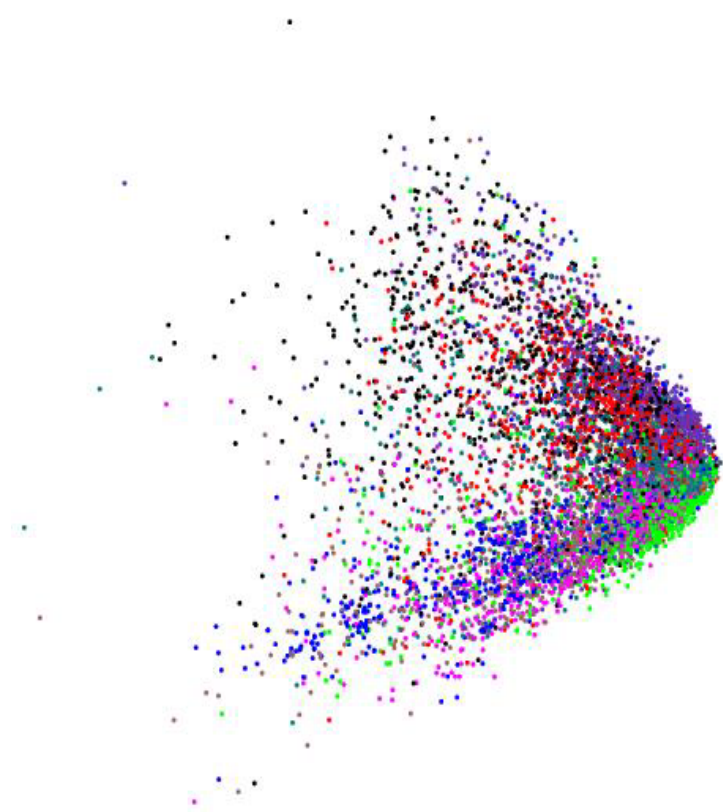
[Data Mining: Practical Machine Learning Tools and Techniques](#) by Ian H. Witten  
★★★★★ (1) £37.04

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# 学什么 What is Learning?

Example 7: Displaying the structure of a set of documents using Latent Semantic Analysis (a form of PCA)

LSA 2-D Topic Space



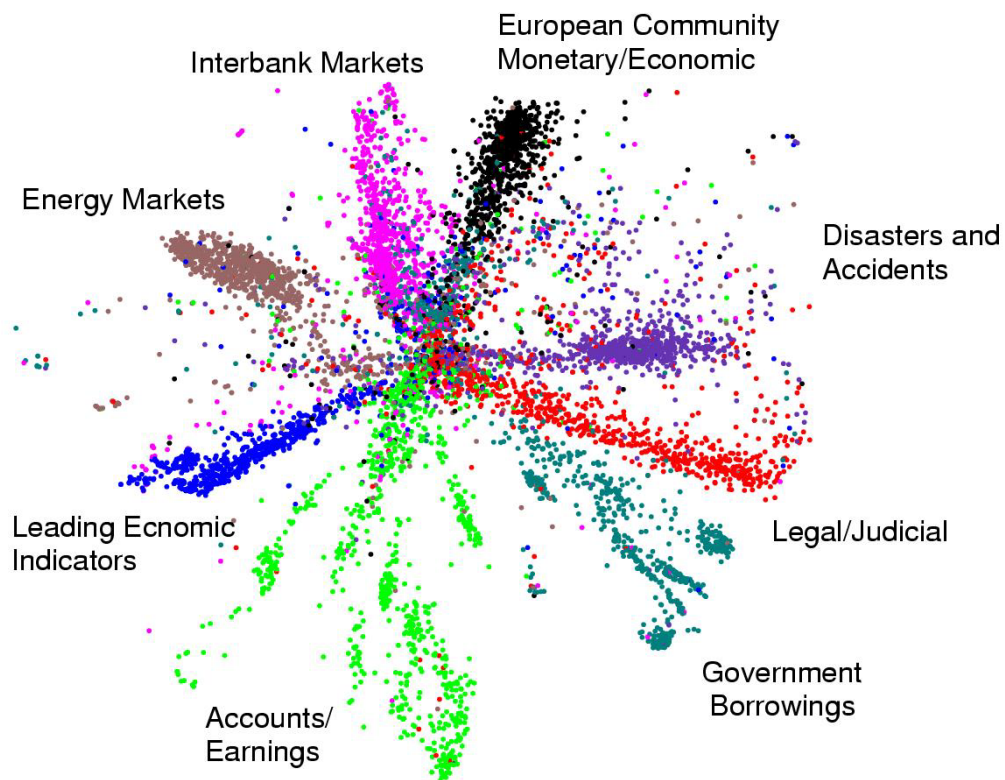
Each document is converted to a vector of word counts. This vector is then mapped to two coordinates and displayed as a colored dot. The colors represent the hand-labeled classes.

When the documents are laid out in 2-D, the classes are not used. So we can judge how good the algorithm is by seeing if the classes are separated.

# 学什么 What is Learning?

Example 7: Displaying the structure of a set of documents using Latent Semantic Analysis (a form of PCA)

Autoencoder 2-D Topic Space



# 学习任务类型

## Types of learning task

- Supervised learning
  - infer a function from labeled training data.
- Unsupervised learning
  - try to find hidden structure in unlabeled training data
  - clustering
- Reinforcement learning
  - To learn a policy of taking actions in a dynamic environment and acquire rewards



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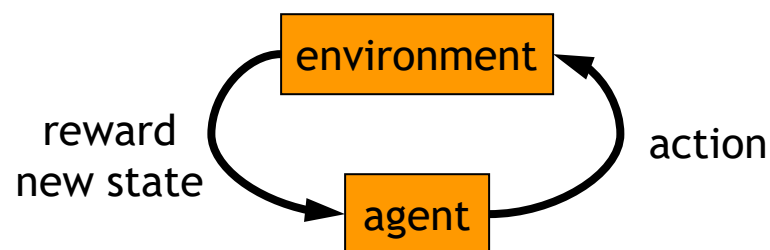




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# 学习任务类型 Types of learning task

## Machine Learning Problems

|                   | <i>Supervised Learning</i>       | <i>Unsupervised Learning</i> |
|-------------------|----------------------------------|------------------------------|
| <i>Discrete</i>   | classification or categorization | clustering                   |
| <i>Continuous</i> | regression                       | dimensionality reduction     |

# 机器学习发展历史

## History of Machine Learning

- 1950s
  - Samuel's checker player
  - Selfridge's Pandemonium
- 1960s:
  - Neural networks: Perceptron
  - Pattern recognition
  - Learning in the limit theory
  - Minsky and Papert prove limitations of Perceptron
- 1970s:
  - Symbolic concept induction
  - Winston's arch learner
  - Expert systems and the knowledge acquisition bottleneck
  - Quinlan's ID3
  - Mathematical discovery with AM

# 机器学习发展历史

## History of Machine Learning

- 1980s:

- Advanced decision tree and rule learning
- Explanation-based Learning (EBL)
- Learning and planning and problem solving
- Utility problem
- Analogy
- Cognitive architectures
- Resurgence of neural networks
- Valiant's PAC (Probably approximately correct) Learning Theory

- 1990s

- Data mining
- Adaptive software agents and web applications
- Text learning
- Reinforcement learning (RL)
- Inductive Logic Programming (ILP)
- Ensembles: Bagging, Boosting, and Stacking
- Bayes Net learning
- Support vector machines
- Kernel methods

# 机器学习发展历史

## History of Machine Learning

- 2000s
  - Graphical models (Bayesian networks and Markov random fields)
  - Variational inference
  - Statistical relational learning
  - Transfer learning
  - Sequence labeling
  - Collective classification and structured outputs
- Computer Systems Applications
  - Compilers
  - Debugging
  - Graphics
  - Security (intrusion, virus, and worm detection)
  - Email management
  - Personalized assistants that learn
  - Learning in robotics and vision

# 机器学习发展历史

## History of Machine Learning

- 2010s
  - Deep learning (DNN, CNN, RNN)
  - ImageNet Challenge
  - Learning from big data
  - Learning with GPUs or HPC
  - Generative Adversarial Networks (GANs)
  - Reinforcement Learning Breakthroughs (AlphaGo)
  - Multi-task & lifelong learning
  - Deep reinforcement learning
  - Natural Language Processing (Transformer, BERT、 GPT )

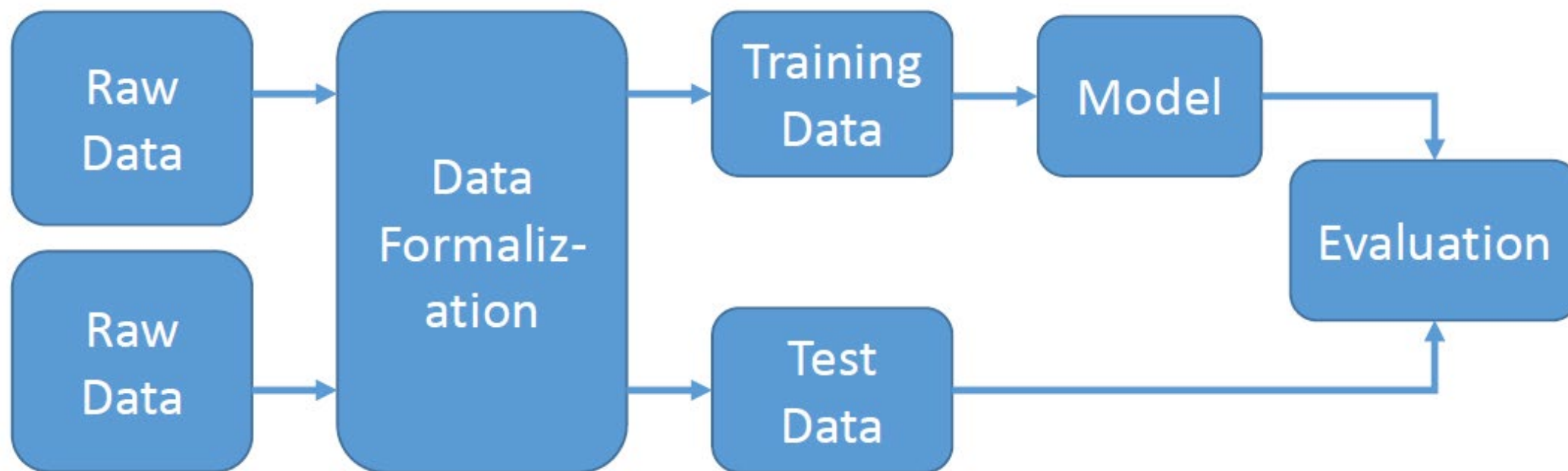
# 机器学习发展历史

## History of Machine Learning

- 2020s...
  - Self-supervised Learning
  - Multimodal Learning (CLIP)
  - Transformer Models (GPT-3、 BERT)
  - Diffusion Models ( DALL-E 2、 Stable Diffusion )
  - Large Language Models (LLMs)
  - Federated Learning
  - Quantum Machine Learning
  - Ethics and Fairness in AI
  - XAI (LIME、 SHAP)
  - ...

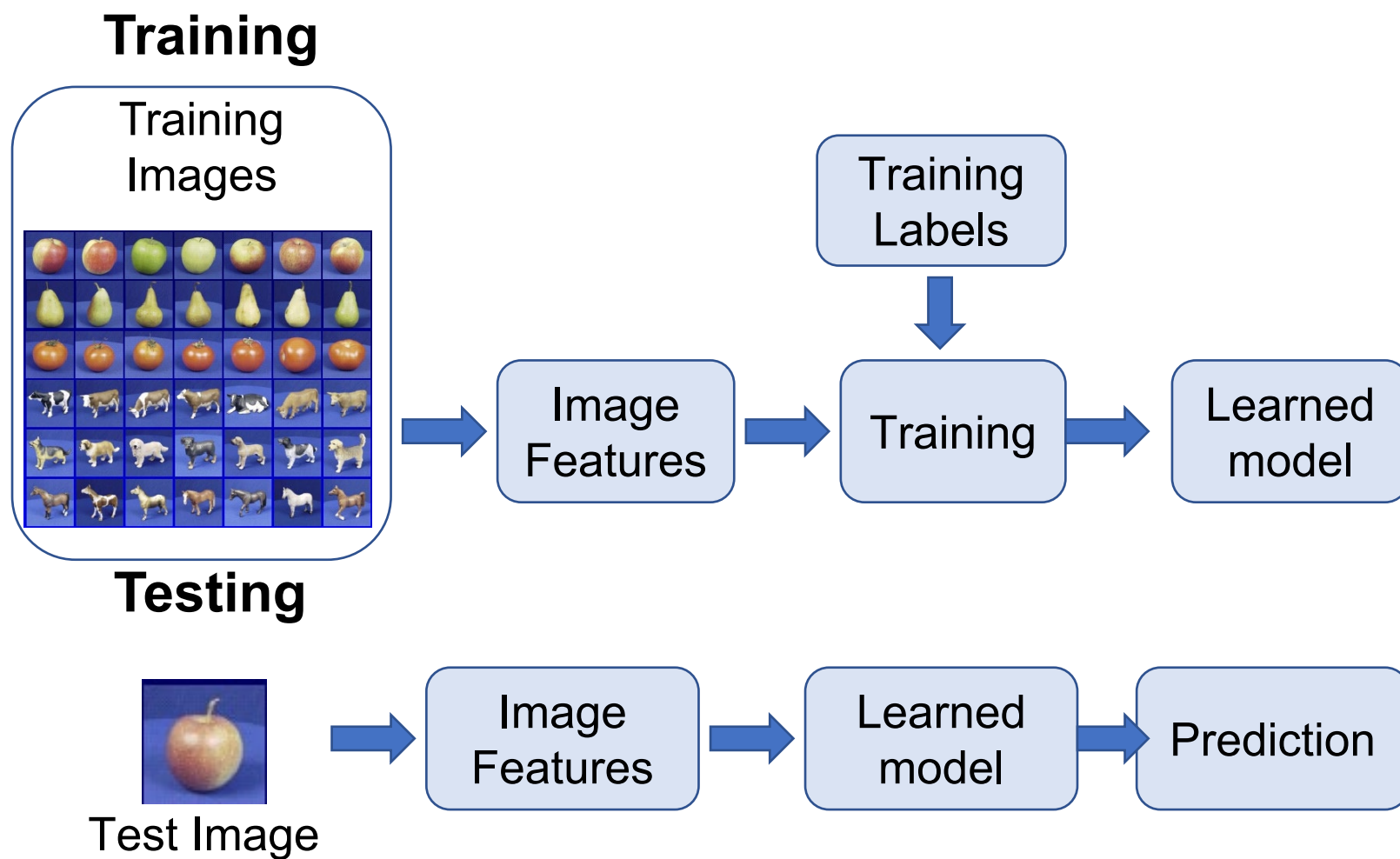


# 机器学习的一般过程 Machine Learning Process



- Basic assumption: there exist the same patterns across training and test data

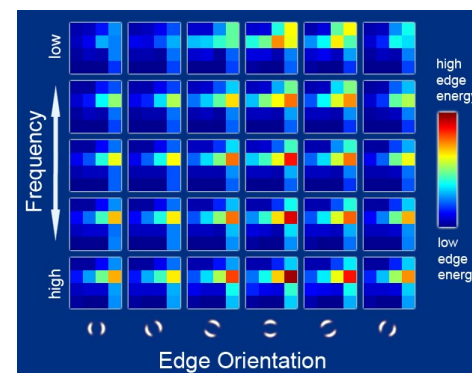
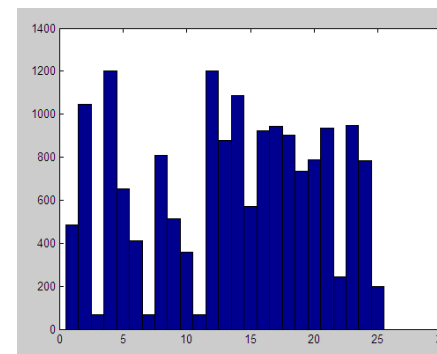
# 图像分类问题举例 Image Classification Example



# 图像分类问题举例 Image Classification Example

## Features

- Raw pixels
- Histogram
- GIST descriptors
- ...



# 图像分类问题举例

## Image Classification Example

- Apply a prediction function to a feature representation of the image to get the desired output:

$f(\text{apple image}) = \text{"apple"}$

$f(\text{tomato image}) = \text{"tomato"}$

$f(\text{cow image}) = \text{"cow"}$

# 图像分类问题举例

Image Classification Example

$$y = f(x)$$

Output label      prediction function      Input instance

- **Training:** given a *training set* of labeled examples  $\{(x^{(i)}, y^{(i)}) \dots, (x^{(N)}, y^{(N)})\}$ , estimate the prediction function  $f$  by minimizing the prediction error on the training set
- **Testing:** apply  $f$  to a never before seen test example  $x$  and output the predicted value  $y = f(x)$

# 监督学习

## Supervised Learning

- Given the training dataset of (data,label) pairs,

$$D = \{(x^{(i)}, y^{(i)})\}_{i=1,2,\dots,N}$$

let the machine learn a function from data to label

$$y^{(i)} \approx f_{\theta}(x^{(i)})$$

- Function set  $\{f_{\theta}(x^{(i)})\}$  is called hypothesis space
- Learning is referred to as updating the parameter  $\theta$  to make the prediction closed to the corresponding label

# 监督学习

## Supervised Learning

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$x^{(i)}$  = input data(features) of  $i^{th}$  training example  
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1. What is the learning objective?
2. How to update the parameters?



# 问题1：学习目标 What is the Learning Objective?

- Make the prediction closed to the corresponding label

(Expected Risk/Error Minimization)

Cannot be directly calculated!

# 问题1：学习目标 What is the Learning Objective?

- Make the prediction closed to the corresponding label

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)}))$$

(Empirical Risk Minimization, ERM)

Loss function  $L(y^{(i)}, f_{\theta}(x^{(i)}))$  measures the error between the label and prediction for single sample.

The definition of loss function depends on the data and task

# 损失函数 Loss function

0-1 Loss Function

$$L(y^{(i)}, f(x^{(i)})) = \begin{cases} 1, & \text{if } y^{(i)} \neq f(x^{(i)}) \\ 0, & \text{if } y^{(i)} = f(x^{(i)}) \end{cases}$$

Mean Squared Error, MSE

$$L(y^{(i)}, f(x^{(i)})) = (y^{(i)} - f(x^{(i)}))^2$$

Absolute Loss Function

$$L(y^{(i)}, f(x^{(i)})) = |y^{(i)} - f(x^{(i)})|$$

Logarithmic Loss Function (Cross-Entropy Loss Function)

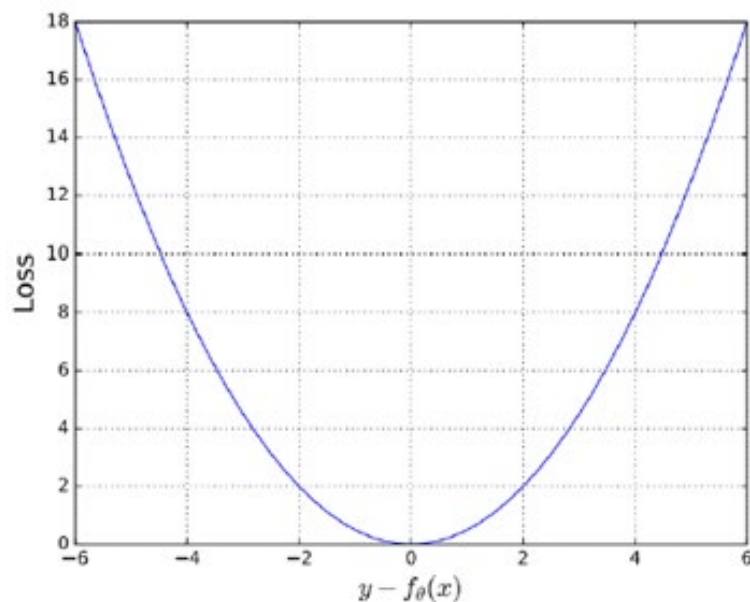
$$L(y^{(i)}, p^{(i)}) = -[y^{(i)} \log(p^{(i)}) + (1 - y^{(i)}) \log(1 - p^{(i)})]$$

$y^{(i)} \in \{0, 1\}$ ,  $p^{(i)} = f(x^{(i)})$  is the predicted probability that the  $i$  th sample belongs to the positive class (usually denoted as class 1))

# 损失函数 Loss function

Most popular loss function: squared loss

$$L(y^{(i)}, f_{\theta}(x^{(i)})) = \frac{1}{2} (y^{(i)} - f_{\theta}(x^{(i)}))^2$$



\*Penalty much more on larger distances

\*Accept small distance (error)  
.Observation  
noise etc.  
.Generalization

## 问题2：如何更新参数 How to update the parameters?

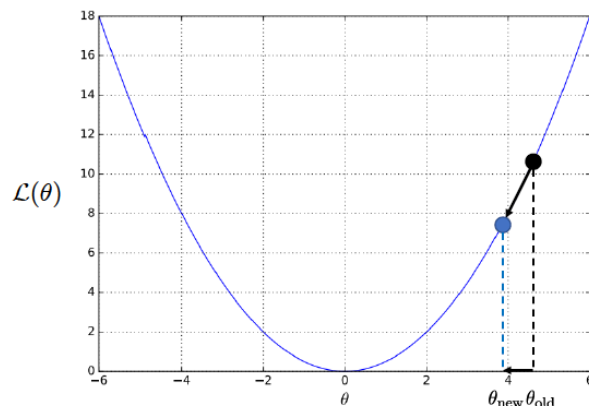
Given  $\theta$ , we have code that can compute

- $J(\theta)$

- $\frac{\partial J(\theta)}{\partial \theta_i}$  (for  $\theta_i \quad j=0, 1, \dots, n$  )

Gradient learning method:

Gradient Learning Methods



$$\theta_{\text{new}} \leftarrow \theta_{\text{old}} - \eta \frac{\partial \mathcal{L}(\theta)}{\partial \theta}$$

Other Optimization algorithms:

- Newton method
- Quasi-Newton method
- Conjugate gradient method

# 超参数 Hyperparameters

Hyperparameters are settings in machine learning models that are set before training begins and are not learned from the data. They can significantly affect model performance.

Key hyperparameters include:

- **Learning Rate:** Controls how quickly the model learns.
- **Regularization Coefficient:** Prevents overfitting by penalizing complex models.
- **Batch Size:** Number of training samples used in one update.
- **Number of Epochs:** How many times the entire dataset is passed through the model.
- **Model Architecture:** Includes the number of layers and neurons in neural networks.
- **Optimizer:** Algorithm used to minimize the loss function.

# 超参数优化方法 Hyperparameter Optimization Methods

- **Grid Search**

Exhaustively explores all possible hyperparameter combinations

- **Random Search**

Randomly samples hyperparameter combinations

- **Bayesian Optimization**

Builds a surrogate model to guide hyperparameter selection

- **Evolutionary Algorithms**

Mimics natural selection to optimize hyperparameters

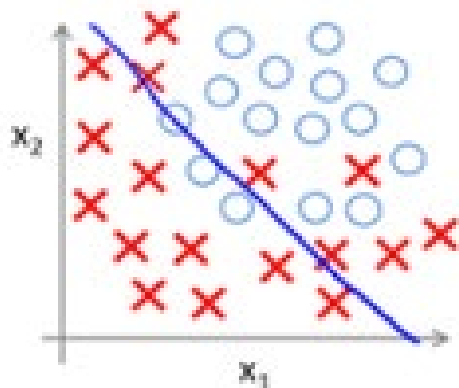
- **Gradient-based Optimization**

Uses gradient information to optimize differentiable hyperparameters

- **Hyperparameter Optimization Libraries**

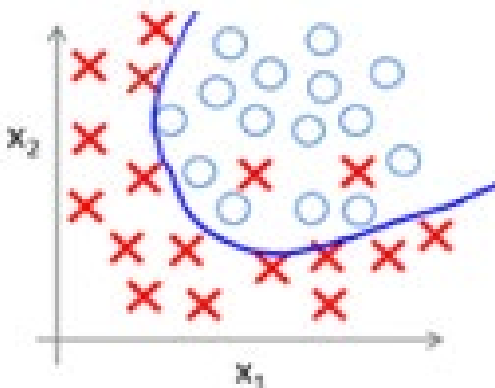
Tools like Optuna, Hyperopt, and Ray Tune support various optimization methods

# 模型选择 Model Selection

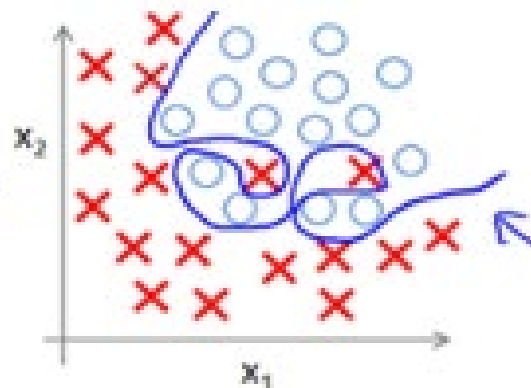


$$f_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

$g = \text{sigmoid function}$



$$f_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2)$$



$$f_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \dots)$$

Linear model: underfitting   Quadratic model: well fitting   Sth.order model: overfitting

- Underfitting occurs when a statistical model or machine learning algorithm cannot capture the underlying trend of the data.
- Overfitting occurs when a statistical model describes random error or noise instead of the underlying relationship



# 模型选择 Model Selection

Trainin  
g data



Test  
data



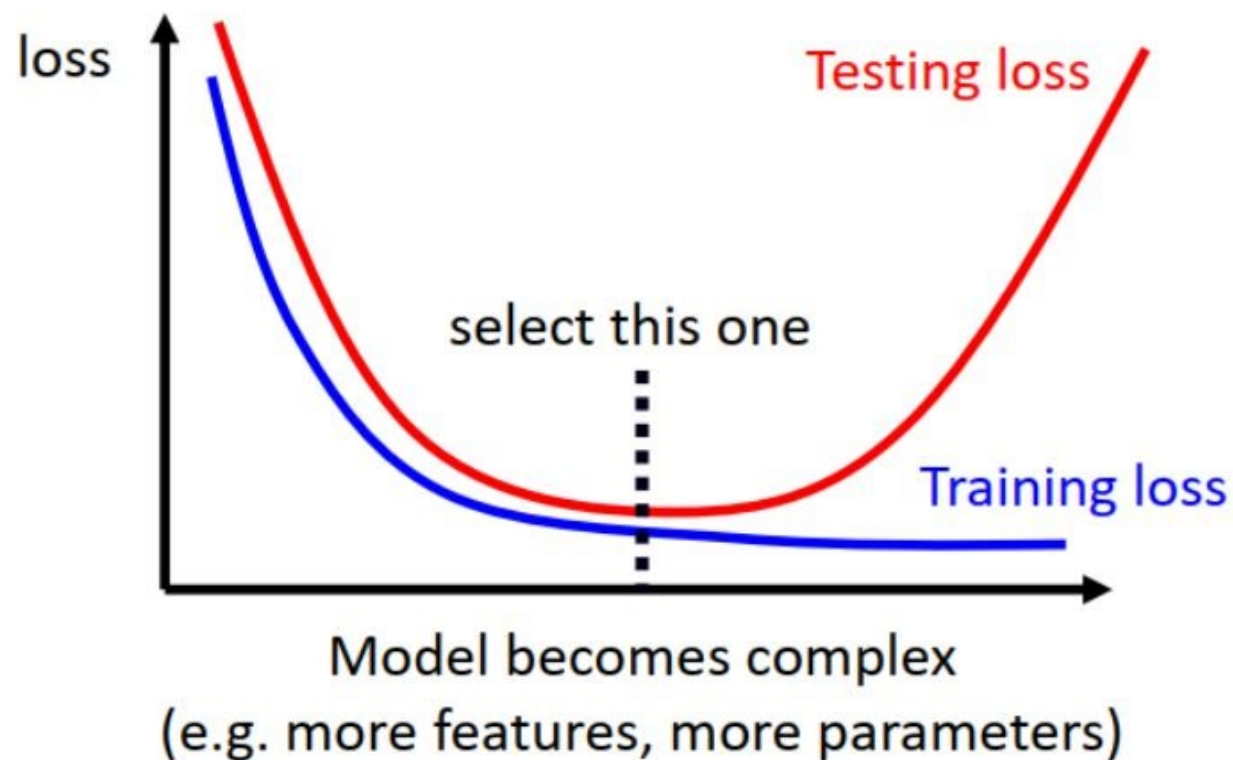
overfitting



underfitting

# 模型选择 Model Selection

## Bias-Complexity Trade-off



# 奥卡姆剃刀原则 Principle of Occam's razor

Among competing hypotheses, the one with the fewest assumptions should be selected.

Recall the function set  $\{f_{\theta}(x^{(i)})\}$  is called hypothesis space

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)})) + \lambda \Omega(\theta)$$

Original loss

Penalty on assumptions

Structural Risk Minimization, SRM

# 正则化方法 Regularization

L2-Norm (Ridge):

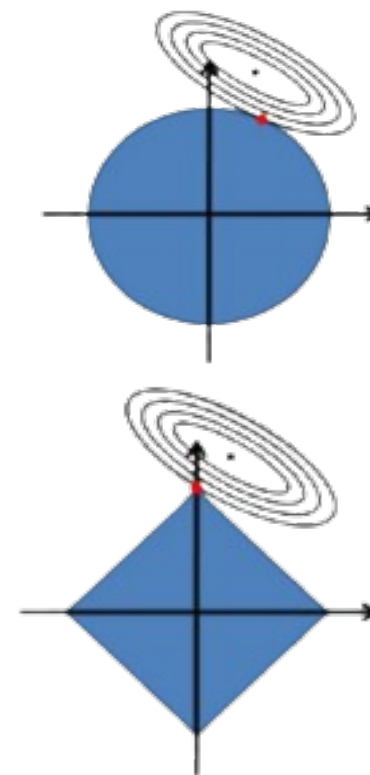
$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)})) + \lambda \|\theta\|^2$$

L1-Norm (LASSO):

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)})) + \lambda |\theta|$$

Elastic Net:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)})) + \lambda |\theta| + (1 - \lambda) \|\theta\|^2$$



# 损失函数&代价函数&目标函数

Loss function & Cost function & Objective function

**Loss Function:**

$$L(y^{(i)}, f_{\theta}(x^{(i)}))$$

**Cost Function:**

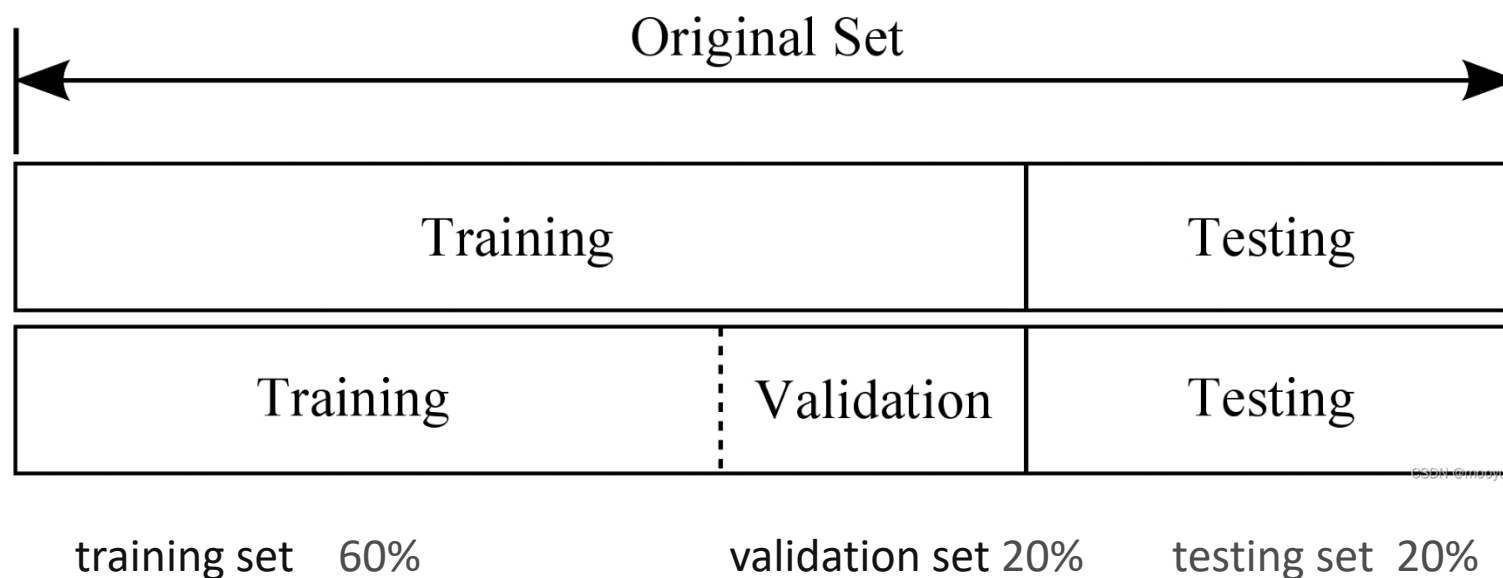
$$\frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)}))$$

**Objective Function:**

The function that the training process aims to minimize or maximize.

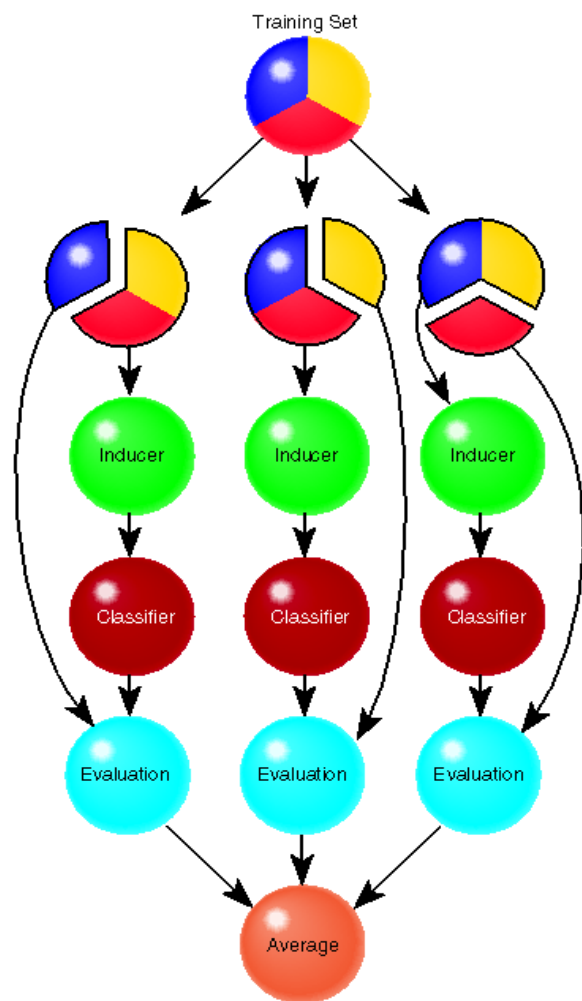
$$\frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)})) + \lambda \|\theta\|^2$$
$$\frac{1}{N} \sum_{i=1}^N L(y^{(i)}, f_{\theta}(x^{(i)})) + \lambda |\theta|$$

# 留出法 hold-out for model selection



Split training data into  
training set and validation set  
for model selection

# 交叉验证 Cross validation for model selection



## K-fold cross validation

```

CV( data  $S$ , alg  $L$ , int  $k$  )
    Divide  $S$  into  $k$  disjoint sets  $\{ S_1, S_2, \dots, S_k \}$ 
    For  $i = 1..k$  do
        Run  $L$  on  $S_{-i} = S - S_i$ 
        obtain  $L(S_{-i}) = h_i$ 
        Evaluate  $h_i$  on  $S_i$ 
         $err_{S_i}(h_i) = 1/|S_i| \sum_{\langle x,y \rangle \in S_i} I(h_i(x) \neq y)$ 
    Return Average  $1/k \sum_i err_{S_i}(h_i)$ 
  
```

leave-one-out cross validation ( $K=|S|$ )

# 偏差和方差 Bias & variance

The true value of  $y$  is given by a function  $f(x)$  plus some noise  $\epsilon$ :  $y = f(x) + \epsilon$



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The mean squared error (MSE) of the model  $\hat{f}(x)$ :  $\epsilon \sim N(0, \sigma^2)$

$$= E[(\hat{f}(x) - y)^2]$$

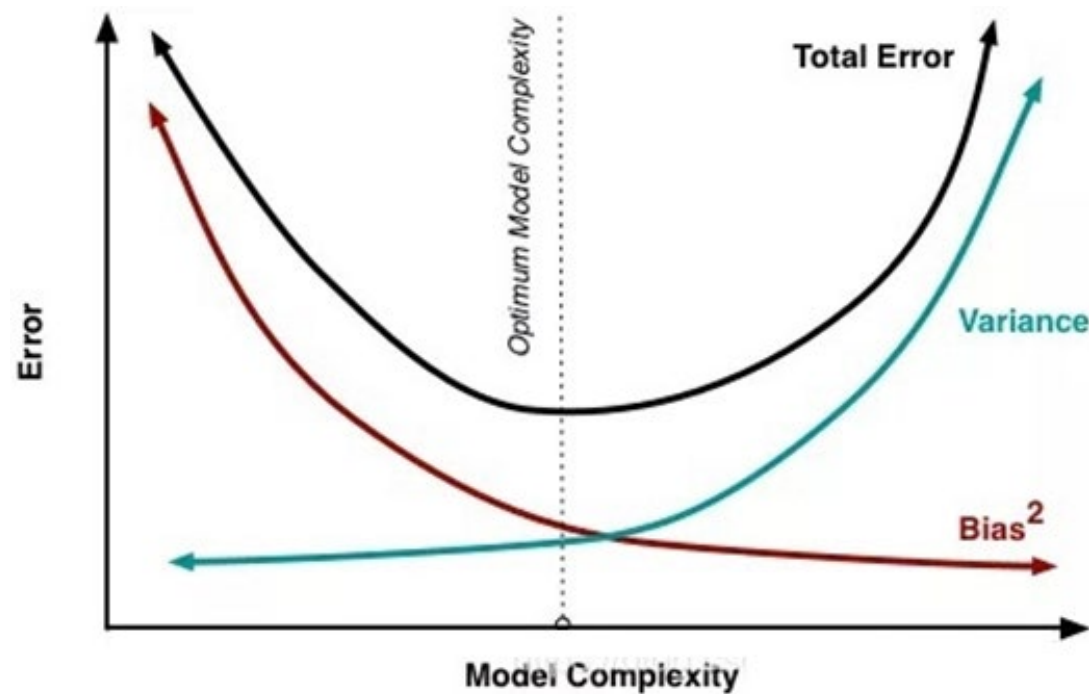
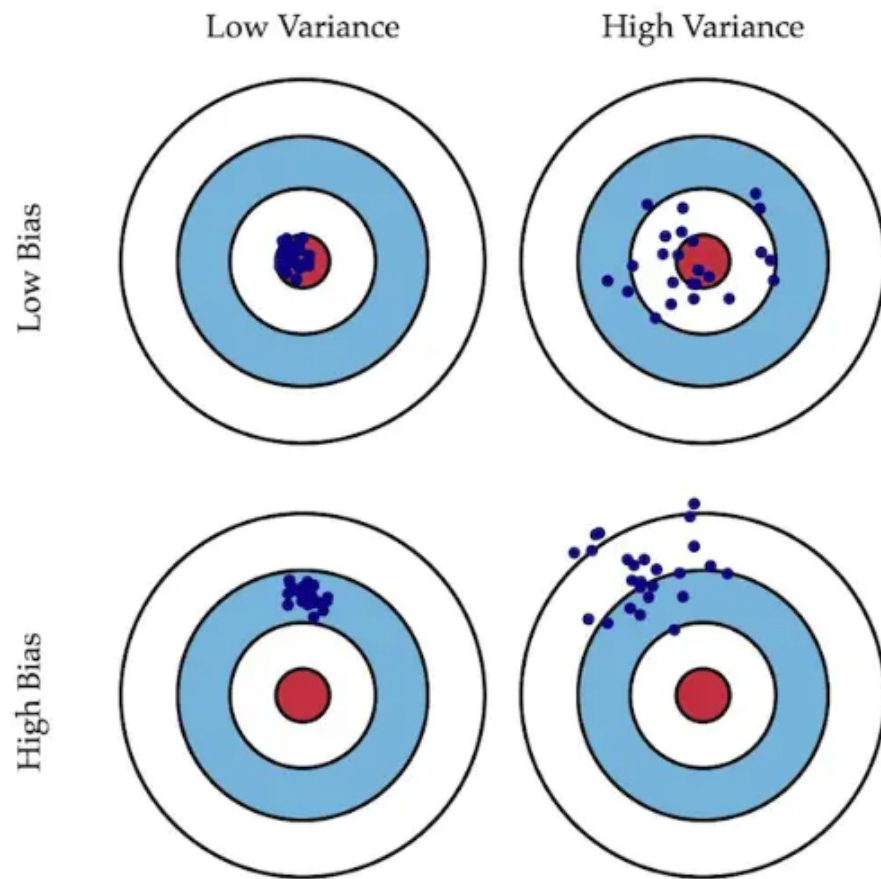
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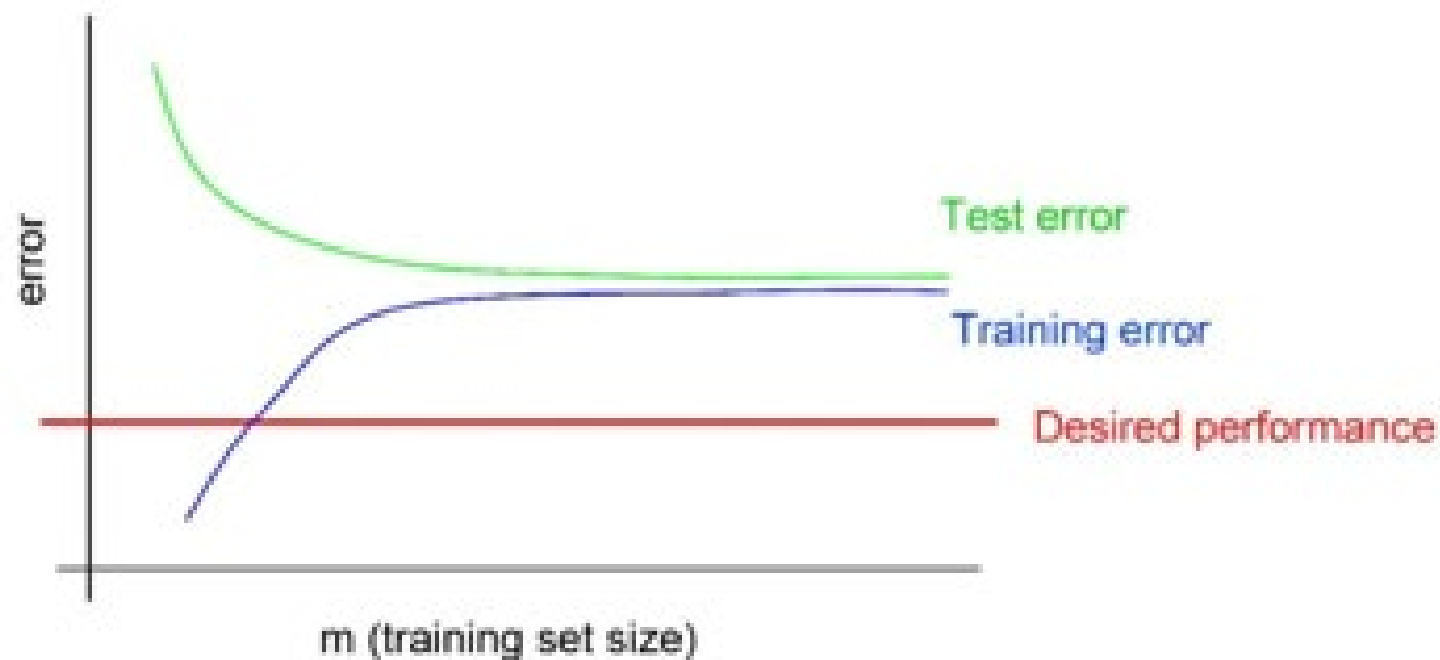
$$\begin{aligned} &= E[(\hat{f}(x) - y)^2] \\ &= E[(\hat{f}(x) - f(x) - \epsilon)^2] \\ &= E[(\hat{f}(x) - f(x))^2] - 2E[(\hat{f}(x) - f(x))\epsilon] + E[\epsilon^2] \\ &= E[(\hat{f}(x) - f(x))^2] + E[\epsilon^2] \\ &= E[(\hat{f}(x) - f(x))^2] + \sigma^2 \\ &= E[(\hat{f}(x) - E[\hat{f}(x)] + E[\hat{f}(x)] - f(x))^2] + \sigma^2 \\ &= E[(\hat{f}(x) - E[\hat{f}(x)])^2] + E[(E[\hat{f}(x)] - f(x))^2] + 2E[(\hat{f}(x) - E[\hat{f}(x)])(E[\hat{f}(x)] - f(x))] + \sigma^2 \\ &= E[(\hat{f}(x) - E[\hat{f}(x)])^2] + E[(E[\hat{f}(x)] - f(x))^2] + \sigma^2 \\ &= \text{Variance} + \text{Bias}^2 + \sigma^2 \end{aligned}$$

# 偏差和方差 Bias & variance



# 偏差和方差 Bias & variance

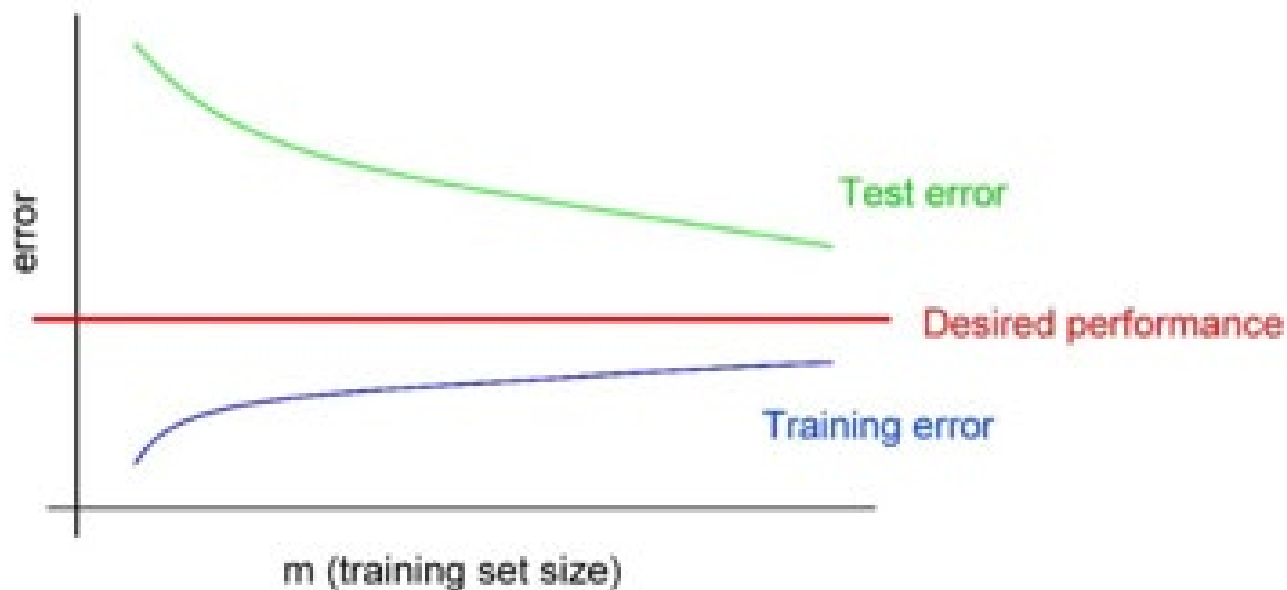
Typical learning curve for high bias:



- Even training error is unacceptably high.
- Small gap between training and test error.

# 偏差和方差 Bias & variance

Typical learning curve for high variance:



- Test error still decreasing as  $m$  increases. Suggests larger training set will help.
- Large gap between training and test error.

# 如何应对欠拟合和过拟合

How to fix underfitting & overfitting

Fixes to underfitting try:

Fixes to overfitting try:

# 如何应对欠拟合和过拟合

How to fix underfitting & overfitting

Fixes to underfitting try:

- Increase model complexity
- Try a larger set of features
- Reduce regularization

Fixes to overfitting try:

- Reduce model complexity
- Try getting more training examples
- Increase regularization
- Try a smaller set of features
- Ensemble learning

# 评价标准

## Evaluation metrics

- For regression tasks:
  - Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean absolute Error (MAE), R-Squared ( $R^2$ )
- For classification tasks:
  - Accuracy, True Positive (TP), False Positive (FP), True Negative (TN), False Negative (FN), ...
- For multi-classification tasks:
  - Take each individual class as the positive class, and the rest as the negative class one by one



# 回归评价标准 Regression Evaluation metrics

MSE(Mean Squared Error)  
均方误差

$$\frac{1}{N} \sum_{i=1}^N (y^{(i)} - f(x^{(i)}))^2$$

MAE (Mean absolute Error)  
平均绝对误差

$$\frac{1}{N} \sum_{i=1}^N |(y^{(i)} - f(x^{(i)}))|$$

RMSE (Root Mean Squared Error)  
均方根误差

$$\sqrt{\frac{1}{N} \sum_{i=1}^N (y^{(i)} - f(x^{(i)}))^2}$$

R-Squared (r2score) R方/决定系数

$$\begin{aligned} &= 1 - \frac{\sum_{i=1}^N (y^{(i)} - f(x^{(i)}))^2}{\sum_{i=1}^N (y^{(i)} - \bar{y})^2} \\ &= 1 - \frac{MSE}{Var} \end{aligned}$$

# 分类性能评价 Classification Evaluation Metrics

Confusion matrix

Predicted classes

Actual  
classes

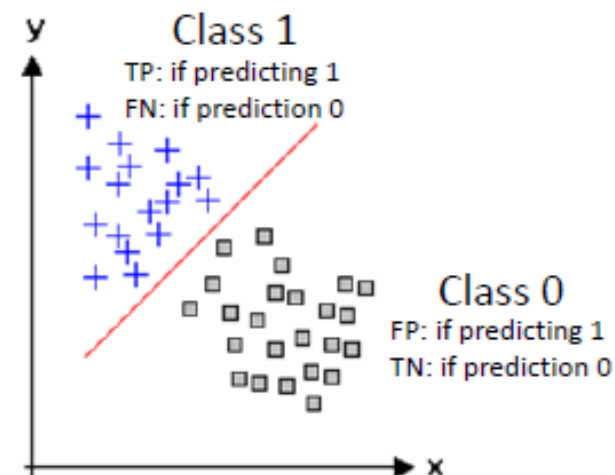
|   | 1              | 0              |
|---|----------------|----------------|
| 1 | True Positive  | False Negative |
| 0 | False Positive | True Negative  |

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$



# 分类性能评价 Classification Evaluation Metrics

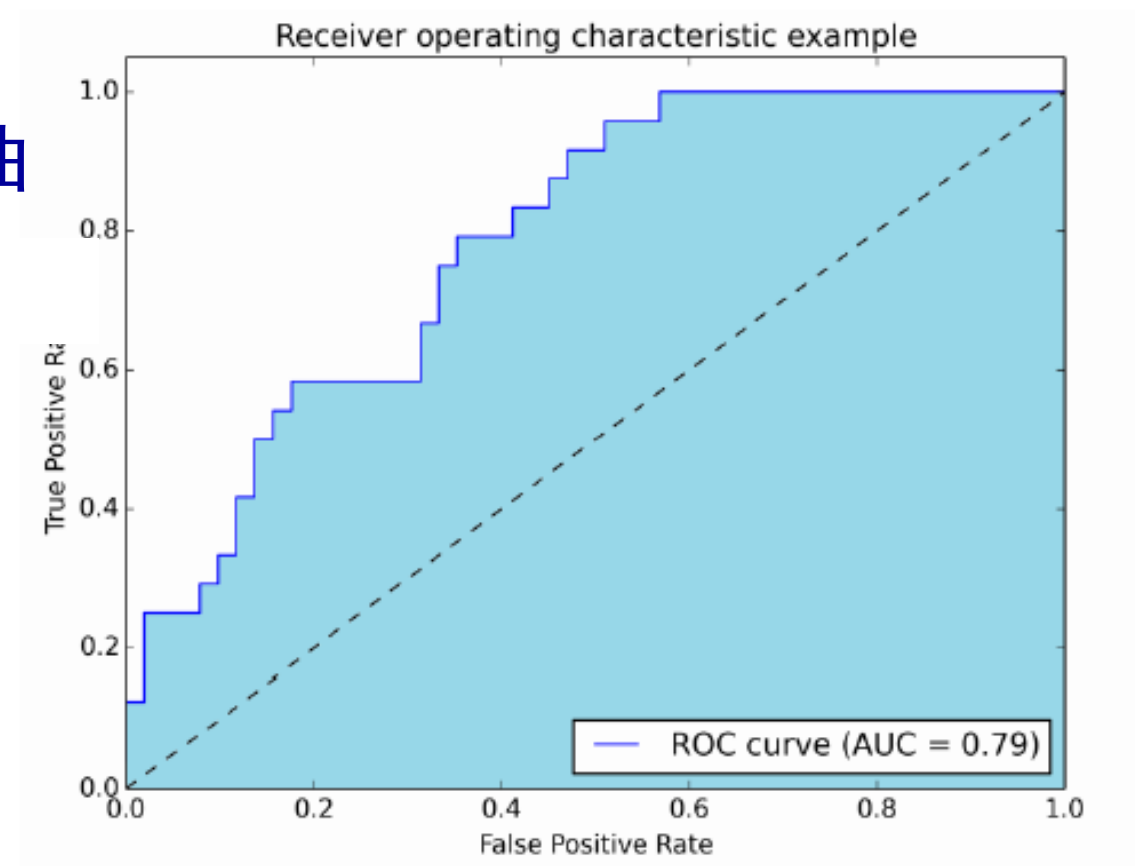
- Ranking-based measure: Area Under ROC Curve (AUC)

分类性能评价：ROC曲线

Receiver Operating Characteristic

True Positive Rate

$$TPR = \frac{TP}{TP + FN}$$

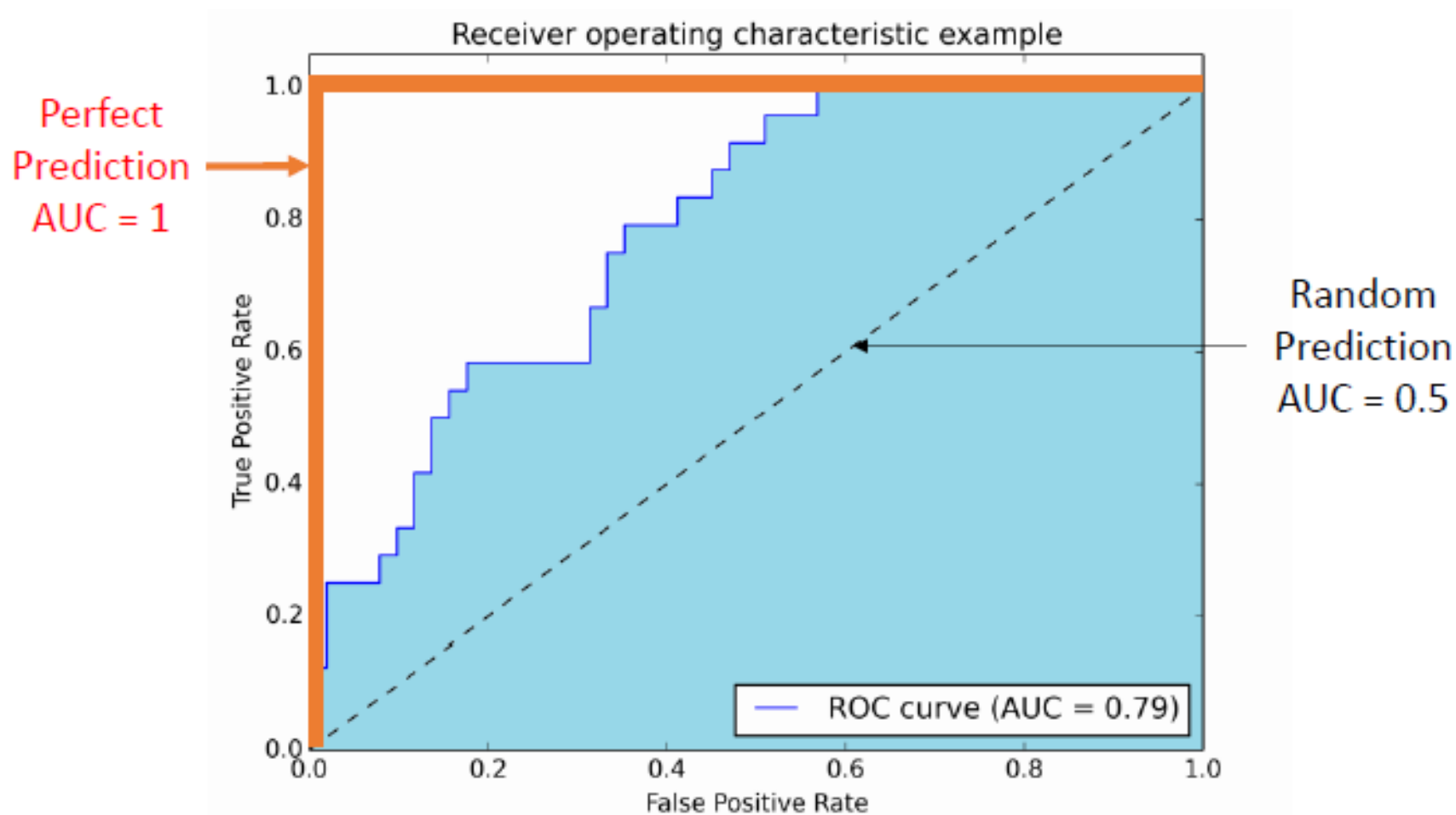


False Positive Rate

$$FPR = \frac{FP}{TN + FP}$$

# 分类性能评价 Classification Evaluation Metrics

- Ranking-based measure: Area Under ROC Curve (AUC)



分类性能评价：  
AUC 面积  
Area under  
Curve

# 监督学习 Supervised Learning

- Given the training dataset of (data,label) pairs,

$$D = \{(x^{(i)}, y^{(i)})\}_{i=1,2,\dots,N}$$

$x^{(i)}$  = input data(features) of  $i^{th}$  training example  
 $y^{(i)}$  = output data(label) of  $i^{th}$  training example

let the machine learn a function from data to label

$$y^{(i)} \approx f_{\theta}(x^{(i)})$$

- Function set  $\{f_{\theta}(x^{(i)})\}$  is called hypothesis space
- Learning is referred to as updating the parameter  $\theta$  to make the prediction closed to the corresponding label

# 机器学习和假设空间 Machine Learning & Hypothesis Space

- One way to think about a supervised learning machine is as a device that **explores a “hypothesis space”**.
- The art of supervised machine learning is in:
  - Deciding how to represent the inputs and outputs
  - Selecting a hypothesis space that is powerful enough to represent the relationship between inputs and outputs but simple enough to be searched.
- **Different learning methods** assume different hypothesis spaces (representation languages) and/or employ different search techniques.

# 函数表征

## Various Function Representations

- Numerical functions
  - Linear regression
  - Neural networks
  - Support vector machines
- Symbolic functions
  - Decision trees
  - Rules in propositional logic
  - Rules in first-order predicate logic
- Instance-based functions
  - Nearest-neighbor
  - Case-based
- Probabilistic Graphical Models
  - Naïve Bayes
  - Bayesian networks
  - Hidden-Markov Models (HMMs)
  - Probabilistic Context Free Grammars (PCFGs)
  - Markov networks

# 搜索算法

## Various Search Algorithms

- Gradient descent
  - Perceptron
  - Backpropagation
- Dynamic Programming
  - HMM Learning
  - PCFG Learning
- Divide and Conquer
  - Decision tree induction
  - Rule learning
- Evolutionary Computation
  - Genetic Algorithms (GAs)
  - Genetic Programming (GP)
  - Neuro-evolution



# 课程安排 Course Arrangement

- Course content :
  1. Linear models (regression)
  2. Linear models (classification);
  3. Decision tree ;
  4. Artificial neural network;
  5. Deep learning;
  6. SVM;
  7. Bayesian Learning
  8. Ensemble learning;
  9. Clustering
  10. Dimension reduction& feature selection
  11. Frontier Introduction
- lab assignments (3人1组) : regression; classification; clustering

# 课程考核 Course Assessment

## ● Class attendance & Quize 15%

- Class quizzes **MUST** be handed to the teacher after each class.

## ● Project (3 times) 45%

- Submission Instructions:

- Submit to: canvas

- Subject: **Student ID\_Name\_MLProject\_X** (replace X with 1, 2, or 3)

- File Name: **Student ID\_Name\_MLProject\_X.zip** (replace X with 1, 2, or 3)

- **Projects Exemption:** Excellent academic paper achievements

## ● Final Exam 40%

# 参考书

- 周志华. “机器学习”. 清华大学出版社, 2016
- 李航, 统计学习方法, 清华大学出版社, 2019
- TOM M MICHELLE. Machine Learning[M]. New York: McGraw-Hill Companies, Inc, 1997.
- Andrew Ng. Machine Learning[EB/OL]. Stanford University, 2014. <https://www.coursera.org/course/ml>
- Bishop 的 Pattern Recognition and Machine Learning, 简称为PRML。中文译本为《模式识别与机器学习》



周志华, 南京大学计算机科学与技术系主任、人工智能学院院长。  
代表作: 《机器学习》(西瓜书)



吴恩达 (Andrew Ng), 斯坦福大学副教授, 前“百度大脑”的负责人与百度首席科学家。



李航, 现任字节跳动科技有限公司人工智能实验室总监, 北京大学、南京大学客座教授, IEEE 会士, ACM 杰出科学家, CCF 高级员。  
代表作: 《统计学习方法》



Christopher M. Bishop, 是机器学习和统计学领域的著名学者之一。

《Pattern Recognition and Machine Learning》被广泛视为机器学习领域的经典教材。





# 预备知识

- ▶ 线性代数
- ▶ 微积分
- ▶ 数学优化
- ▶ 概率论
- ▶ 信息论

<https://nndl.github.io/>  
《数学基础》

# Resources: Datasets

- 通用性分析：
  - UCI Repository: <http://www.ics.uci.edu/~mlearn/MLRepository.html>
  - UCI KDD Archive: <http://kdd.ics.uci.edu/summary.data.application.html>
  - Statlib: <http://lib.stat.cmu.edu/>
  - Delve: <http://www.cs.utoronto.ca/~delve/>
- 探索性分析： Kaggle数据集： <https://www.kaggle.com/datasets>
- 深度学习：
  - [Deeplearning.net](http://Deeplearning.net) - 用于对深度学习算法进行基准测试的最新数据集列表
  - [DeepLearning4J.org](http://DeepLearning4J.org) - 用于深度学习研究的高质量数据集的最新列表
  - <https://paperswithcode.com/datasets>

# Resources: Journals

- Journal of Machine Learning Research [www.jmlr.org](http://www.jmlr.org)
- Machine Learning
- Neural Computation
- Neural Networks
- IEEE Transactions on Neural Networks
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- Annals of Statistics
- Journal of the American Statistical Association
- ...

# Resources: Conferences

- International Conference on Machine Learning (ICML)
- Neural Information Processing Systems (NIPS→NeurIPS)
- Computational Learning Theory (COLT)
- European Conference on Machine Learning (ECML)
- Asian Conference on Machine Learning (ACML)
- IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
- AAAI Conference on Artificial Intelligence, AAAI) :
- 1 (International Joint Conference on Artificial Intelligence, IJCAI)
  
- 中国机器学习大会(CCML)
- 机器学习及其应用 (MLA)





# 问题

- 试分析什么因素会导致模型出现图中所示的高偏差和高方差情况。

